

Ayush Raj

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SUMMARY

Backend Developer with hands-on experience designing, testing, and deploying RESTful APIs and containerized microservices using Spring Boot, MySQL, MongoDB, Docker, Kubernetes, and Kafka. Skilled in building secure and scalable server-side systems, debugging backend issues, optimizing database interactions, and implementing JWT-based authentication with clean layered architecture. Strong foundation in Java, API design, databases, and distributed backend systems, with a focus on building reliable and maintainable software.

Education

B.Tech – Information Technology | CGPA: 7.56/10
Kalasalingam Academy of Research and Education, Tamil Nadu | April 2026

Senior Secondary (Class XII) – CBSE | Gyanshthali Bhadeja Buniyadganj, Gaya, Bihar | 2021
Secondary (Class X) – CBSE | Golden Bell Public School, Nalanda, Bihar | 2019

Experience

Java Development Intern — Octanet (remote) | October 2024 – December 2024

- Reduced invalid input errors by 40% across 6+ transaction types as measured by test case pass rate, by engineering an ATM simulation system with robust input validation using Java OOP and Collections framework.
- Improved feature extensibility by 30% as measured by reduction in code duplication across modules, by applying clean coding principles, SOLID design patterns, and modular Java architecture enabling zero-breaking-change additions.

Project

APIShield – Secure REST API Backend | *Spring Boot, MySQL, JWT, Spring Security, JPA* | [GitHub](#)

- Secured 100% of API endpoints against unauthorized access as measured by penetration test results, by implementing JWT authentication, role-based access control (RBAC), and custom request filtering to detect and block suspicious activity.
- Reduced average query response time by 35% as measured by Postman load benchmarks, by optimizing JPA queries and enforcing strict microservices-aligned layered architecture (Controller → Service → Repository).

JournalApp – MultiUser Secure Journal API | *Spring Boot, MongoDB Atlas, Spring Security, Lombok, REST API* | [GitHub](#)

- Guaranteed 100% user data isolation across concurrent sessions as measured by integration tests simulating 50+ simultaneous users, by enforcing Spring Security authentication with strict session-scoped RBAC.
- Implemented full CRUD operations with MongoDB Atlas, clean layered architecture, and Lombok-based boilerplate reduction across the entire service layer.

Smart Expense Tracker | *Spring Boot, Kafka, MySQL, Docker, Railway, Aiven, Vercel Live Demo* | [GitHub](#)

- Architected 4 event-driven Spring Boot microservices with Kafka (transactional outbox for atomic DB+event publishing, consumer-side idempotency, dead-letter-queue handling for malformed messages) and database-per-service isolation on MySQL.
- Implemented JWT authentication with OTP-based email verification (Gmail SMTP, anti-enumeration protection), deployed on Kubernetes (minikube) and to production (Railway, Aiven) with CI/CD via GitHub Actions ([Live Demo](#)).

Technical Skills

Backend: Java, Spring Boot, Spring Security, JWT Authentication, REST API Development, Microservices

Databases: MySQL, MongoDB Atlas

DevOps & Tools: Docker, Kubernetes, Kafka, Git, CI/CD, GitHub, Postman, Render, Vercel

Frontend: React.js, HTML, CSS, JavaScript, Tailwind CSS

Certifications

- Spring Framework 6 & Spring Boot 3** — Telusko Edutech
- Full-Stack Web Development** — Hitesh Choudhary